

Summary of SUAS 2023

Virginia-Tech Submission

University Of Virginia

Imaging System

The team opted for a straightforward, manual approach to their imaging system, prioritizing reliability and minimizing processing overhead. The imaging system utilizes a GoPro Hero 7 Black and the camera is connected to a Herelink Air Unit, which transmits a real-time video feed to the Ground Control Station (GCS) at 4K resolution and 60 frames per second (fps).

They are using Manual ODLC Instead of using complex onboard image processing algorithms for Object Detection, Classification, and Localization (ODLC), the team streams the video down to the ground operators. The operators manually handle the detection, classification, and localization of targets. The logic behind this being that this manual process, while requiring an additional operator, significantly reduces the probability of a failed airdrop attempt that could be caused by poor algorithm performance.

Before integrating the GoPro, they tested image resolutions using a DJI Mavic Air at various altitudes. Through testing, they determined that hovering at approximately 80 feet provided the highest image quality, achieving a resolution of about 2.5 centimeters per pixel.

Communications Approach

The communication system is built around consolidating functions to ensure reliability and range. The team uses the Herelink Air Unit and Pilot Controller system, this connects both the Cube Orange autopilot (for telemetry) and the GoPro Hero 7 (for video). This single compact device combines the telemetry transmitter and ground control functions. Because the team offloaded all of the imaging and localization work to the human operators on the ground, they do not use a dedicated companion onboard computer for processing. The drone's "brain" is strictly limited to flight control, utilizing a Cube Orange autopilot running open-source ArduCopter software, paired with a Here 3 GPS module.

The Herelink Air Unit transmits both the high-definition video and the telemetry data over a single 2.4 GHz frequency band, It has an estimated range of over 12 miles (with claims of higher ranges using directional antennas) and is noted for reliable performance with (less than 5% packet losses) even around structures or obstructions.

The GCS (Receiver) On the ground, all communication between the drone (Jaeger) and the ground flows directly to the Herelink system. To complete the GCS setup, this controller is then connected via a wired connection to a laptop running QGroundControl (QGC). This mirroring ensures the

ground operators have a highly reliable, large-screen interface to monitor the drone and perform manual localization without relying solely on the handheld controller's screen.